



SARCASM DETECTOR

AI-ML MODEL PROJECT

Abstract

Course Name: Introduction to AI-ML

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Sarcasm Detector: Project Report

1. Project Approach

Our goal was to build a machine learning model to detect sarcasm in headlines, as per the "Sarcasm Detector" problem statement. The dataset was relatively small (~28,000 headlines), so our strategy focused heavily on Feature Engineering rather than complex deep learning models.

Our approach consisted of a hybrid feature set:

- . Linguistic Features (TF-IDF): To capture *what* is being said.
- . Meta-Features: To capture *how* it's being said (e.g., exaggeration, punctuation).

We used a `scikit-learn Pipeline` to combine these features, train two candidate models (Logistic Regression and SVM), and select the best-performing one using 5-fold cross-validation.

2. Features Used

Our final model was trained on a combination of two feature sets:

A. Text Features (TF-IDF)

- Vectorizer: `TfidfVectorizer`
- N-grams: `(1, 2)` - This captures both single words (e.g., "great") and two-word phrases (e.g., "yeah right"), which is crucial for understanding sarcastic context.
- Parameters: We removed English "stop words" and limited the vocabulary to the top 5,000 features to prevent overfitting.

B. Engineered Meta-Features We engineered four new features to quantify the *style* of the headline:

- `exclamation_count`: The number of exclamation marks, often a sign of emphasis or mock excitement. `question_count`: The number of question marks, often used in rhetorical sarcastic questions. `all_caps_count`: The number of words written in ALL CAPS (e.g., "WOW," "GREAT"), a strong indicator of exaggeration. `word_count`: The total number of words in the headline.
 - These numerical features were scaled using `StandardScaler` to be on the same level as the TF-IDF features.
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3. Model Choice

We evaluated two models suitable for this type of classification task: Logistic Regression and Linear Support Vector Machine (LinearSVC).

To get a reliable performance estimate, we used 5-fold cross-validation on our training data.

Cross-Validation Results:

- Logistic Regression Mean Accuracy: 0.8872 Linear SVM
- Mean Accuracy: 0.9015

The Linear SVM consistently outperformed Logistic Regression in cross-validation. Therefore, we selected the `LinearSVC` as our final model.

The final, trained model achieved an accuracy of [Check your output from Cell 7, e.g., 0.9034] on the held-out test set.

4.Group Activities

